6TH GRADE MATH · DATA DISPLAYS

Dot Plot Balance: Mean, Median & Spread

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Dot Plot Balance: Mean, Median & Spread** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. Data: 5, 6, 6, 7, 8, 8, 9, 10, 13. If 13 changes to 20, which changes more, the mean or the median?
   The mean (8 becomes about 8.78). The median stays 8.
2. Where would you put a fulcrum (▲) under a dot plot so it balances?
   At the mean.

2 Explore use the tool

1. Record the mean and median of the starting data.
   Mean 8, median 8.
2. Drag 13 to 20. Record the new mean and median.
   Mean $\frac{79}{9} \approx 8.78$; median 8.
3. Show the distances from the mean. Record the left and right totals for the starting data. Try another data set. What do you notice?
   8 and 8. The totals are always equal.
4. Make two data sets with mean 8: one with MAD less than 1 and one with MAD more than 3. Describe each dot plot.
   Low MAD: dots bunched near 8. High MAD: dots spread far from 8. (Data sets vary.)

3 Record use the tool and fill in the table

Build each data set (Remove values until 5 are left, then drag)

Quartiles: the median of each half, leaving out the middle value.

Data	Mean	Median	Range	IQR	MAD
4, 6, 8, 10, 12	8	8	8	6	2.4
8, 8, 8, 8, 8	8	8	0	0	0
2, 8, 9, 9, 12	8	9	10	5.5	2.4
1, 2, 2, 3, 20	5.6	2	19	10	5.76

4 Explain what did you discover?

1. The mean is the balance point: the total distance below it equals the total distance above it.
2. An outlier pulls the mean toward it more than the median.

3. MAD is the average **distance** from the mean.

4. Two classes both have a mean quiz score of 8. Class A has a MAD of 0.5 and Class B has a MAD of 3. What does that tell you about the two classes?
   Class A's scores are all close to 8. Class B's scores are much more spread out: a typical score is about 3 points from 8.

5 Apply a new situation, no tool

1. Find the mean and median of 3, 4, 4, 5, 34. Which one better describes a typical value? Why?
   Mean 10, median 4. The median, because 34 is an outlier that pulls the mean up.
2. The mean of 4 numbers is 6. Three of them are 2, 5 and 9. Find the fourth.
   8 (the total must be 24).
3. Find the MAD of 2, 4, 6, 8.
   Mean 5; distances 3, 1, 1, 3; MAD = 2.

Teacher notes

Intended discovery: The mean is the balance point (distances on each side are equal), the median depends only on order, and MAD measures typical distance from the mean.

Likely misconceptions:

- Thinking the mean and median always move together.
- Treating MAD or range as a measure of center.
- Computing quartiles by including the median (this tool and the key use the exclusive method).

Answers that can vary: Explore 4 data sets and most Try this answers (for example, any 9 values with sum 90 and no 10 for "mean exactly 10").